

COMP30027 Report

1 Introduction

This project studies supervised image classification under a coarse-to-fine setting. Task 1 is a coarse-grained animal classification problem, where the target classes belong to visually distinct animal categories. Task 2 is a fine-grained bird species classification problem, where the classes are more visually similar and often differ only in subtle colour, texture, shape, or local markings. The contrast between these two tasks makes it possible to examine how feature representations and classifiers behave as the classification problem becomes more fine-grained.

In this report, I compare provided hand-crafted image descriptors, including colour histograms, HOG, and additional low-level features, with learned representations extracted from ImageNet-pretrained convolutional neural networks. I then evaluate several classifiers on both tasks using accuracy and macro-F1, analyse the main performance differences, and discuss the errors and limitations revealed by the coarse-to-fine progression.

2 Methodology

2.1 Feature Representations

For both tasks, I first used the provided hand-crafted image descriptors, including colour histograms, HOG descriptors, and additional low-level image statistics. These representations capture colour distribution, edge orientation, texture variation, and simple global image properties. They provide interpretable baselines, although they may be limited for fine-grained recognition where class differences are subtle and local. HOG descriptors were included because edge orientation is a useful cue for shape and local structure (Dalal and Triggs, 2005).

I also used ImageNet-pretrained convolutional neural networks as fixed feature extractors. In this approach, each image is passed through a pretrained CNN, and the final classification layer is removed so that the network output is used as a deep embedding. I experimented with **ResNet-50** and **EfficientNet** variants, with EfficientNet-V2-L giving the strongest representation in later experiments.

These CNN outputs were treated as engineered feature representations rather than end-to-end neural classifiers, so they could be concatenated with the provided features and passed to the same set of supervised learners. The CNNs were not fine-tuned on the project data, CIFAR-10, or CUB-200-2011; they were used only as generic ImageNet-pretrained feature extractors, which satisfies the project constraints (Deng et al., 2009; Tan and Le, 2021).

2.2 Classifiers

I evaluated linear, non-linear, probabilistic, distance-based, and ensemble classifiers. Logistic Regression and Linear SVM tested whether the learned feature space was approximately linearly separable, while RBF SVM allowed curved decision boundaries (Cortes and Vapnik, 1995). I also evaluated KNN, Gaussian Naive Bayes, Random Forest, and Extra Trees; the tree ensembles can capture non-linear feature interactions (Breiman, 2001; Geurts et al., 2006).

2.3 Evaluation Protocol

For model evaluation, I used stratified train-validation splits for the main result tables and error analysis, and 5-fold stratified cross-validation during hyperparameter screening. Thus, the validation results reported in the tables refer to the held-out validation split unless explicitly described as cross-validation. Stratification was used to preserve the class distribution in each split, which is important for both 10-class classification tasks. Accuracy was the primary metric because the Kaggle competitions evaluate submissions using classification accuracy. Macro-F1 was also reported to check whether performance was balanced across all classes rather than dominated by easier categories.

Hyperparameters were selected using only the labelled training data, including regularisation strength, preprocessing strategy, SVM kernel parameters, and tree ensemble settings. After selecting the best configuration, the final model was retrained on the full labelled training set before generating Kaggle predictions for the unlabelled test images.

3 Results

3.1 Task 1: Coarse-Grained Classification

Table 1 summarises the main Task 1 validation results. Using only the provided hand-crafted descriptors gave relatively weak performance, with the best baseline accuracy around 54.4%. Adding ImageNet-pretrained CNN embeddings substantially improved performance. The best Task 1 validation result was obtained using colour, HOG, additional features, and EfficientNet-V2-L embeddings, reaching 97.3% accuracy and 97.3% macro-F1 with Logistic Regression. This suggests that the pretrained CNN representation captured more discriminative visual structure than simple hand-crafted features alone.

Features	Accuracy	Macro-F1
Colour + HOG + additional	0.544	0.541
Colour + HOG + additional + ResNet-50	0.897	0.897
HOG + additional + EffNet-V2-M	0.931	0.931
Colour + HOG + additional + EffNet-V2-L	0.973	0.973

Table 1: Selected Task 1 held-out validation results. EffNet denotes EfficientNet.

After selecting the strongest Task 1 representation, I compared several classifiers on the same colour, HOG, additional features, and EfficientNet-V2-L feature set. Table 2 shows a small comparison across linear, kernel-based, and ensemble models.

Model	Accuracy	Macro-F1
Logistic Regression	0.973	0.973
RBF SVM	0.963	0.963
Random Forest	0.960	0.960

Table 2: Selected Task 1 held-out validation comparison of models using colour, HOG, additional features, and EfficientNet-V2-L embeddings.

Logistic Regression performed best, while RBF SVM and Random Forest were close. This suggests that the EfficientNet-V2-L representation made Task 1 largely separable without requiring a more complex classifier.

3.2 Task 2: Fine-Grained Classification

Task 2 was more challenging than Task 1 because the classes are visually similar bird species

and the labelled training set is much smaller. Initial experiments using only the provided low-level descriptors gave limited performance, with the best provided-feature model reaching 40.5% accuracy. Adding CNN embeddings produced a much larger improvement, as shown in Table 3. The strongest representation combined HOG, additional features, and EfficientNet-V2-L embeddings, and this feature set gave consistently strong performance across different classifiers.

Feature set (best model)	Acc.	Macro-F1
Colour + HOG + additional, Random Forest	0.405	0.392
HOG + additional + ResNet-50, Extra Trees	0.845	0.838
HOG + additional + EffNet-V2-L, Random Forest	0.976	0.975

Table 3: Task 2 held-out validation comparison between provided low-level features and the final deep-feature representation.

Table 4 reports the main Task 2 results for this strongest representation. Tree-based ensemble models achieved the strongest performance, with Random Forest and Extra Trees obtaining the highest accuracy among the evaluated classifiers. Logistic Regression and RBF SVM also performed strongly, which suggests that the EfficientNet-V2-L embedding made the bird species substantially more separable than the hand-crafted descriptors alone. The relatively small gap between the best models also indicates that the representation itself was more important than the choice of classifier.

Model	Accuracy	Macro-F1	Notes
Random Forest	0.976	0.975	Ensemble
Extra Trees	0.976	0.975	Ensemble
Logistic Reg.	0.952	0.951	Linear
RBF SVM	0.952	0.951	Kernel
Linear SVM	0.952	0.951	Linear
KNN	0.940	0.939	Distance
Gaussian NB	0.940	0.938	Probabilistic

Table 4: Task 2 held-out validation results using HOG, additional features, and EfficientNet-V2-L embeddings.

3.3 Cross-Task Comparison

Across both tasks, the strongest results came from combining selected provided descriptors with ImageNet-pretrained CNN embeddings. The final feature choices also reflected the coarse-to-fine setting. Colour was kept for Task 1 because coarse animal categories often differ in global colour and overall appearance. For Task 2, the global colour histogram was not included in the final feature set because bird species can have highly overlapping colour dis-

tributions; it was therefore more likely to be redundant or noisy than HOG, additional features, and EfficientNet-V2-L representations, which better encode shape, texture, and local visual evidence. Once EfficientNet-V2-L embeddings were included, several classifiers achieved similar validation performance, indicating that the main gain came from improving the representation rather than only choosing a more complex classifier.

4 Discussion and Critical Analysis

4.1 Effect of Feature Representations

The main finding is that the choice of representation had a larger effect than the choice of classifier. The provided descriptors were useful baselines because they encode interpretable low-level properties such as colour distribution, edge orientation, texture, and simple image statistics. However, these features do not fully capture invariance to pose, viewpoint, scale, and background variation. This explains why they were not sufficient on their own, especially for Task 2.

Pretrained CNN embeddings improved both tasks because they provide a hierarchical visual representation learned from a much larger image collection (Deng et al., 2009; Yosinski et al., 2014). Earlier layers capture local edges and textures, while later layers represent more abstract object parts and shapes (Zeiler and Fergus, 2014). This is useful for Task 1 because animal categories can often be separated by broad shape and appearance cues. It is even more important for Task 2, where the relevant evidence may be a small local cue such as a bill marking, head colour, or wing pattern. In this sense, the CNN features reduce the burden placed on the classifier: after the image has been mapped into a stronger feature space, even relatively simple models can separate many classes well.

The improvement from ResNet-50 and EfficientNet-V2-M to EfficientNet-V2-L suggests that representation capacity also mattered. The stronger model likely captured richer visual patterns that transferred better to fine-grained bird recognition. At the same time, the CNNs were used only as fixed ImageNet-pretrained feature extractors, not fine-tuned on CUB-200-2011 or the project data. Therefore, the performance gain should be interpreted as the benefit of transfer learning rather than supervised adaptation to the target classes.

4.2 Effect of Classifier Choice

Classifier choice still mattered, but its effect was smaller once strong CNN embeddings were included. Logistic Regression and Linear SVM performed well, which suggests that EfficientNet-V2-L made many classes close to linearly separable. RBF SVM did not give a large improvement over the linear models, which again indicates that much of the useful non-linearity had already been introduced by the CNN feature extractor.

This connects to the bias-variance trade-off discussed in lectures. With a high-dimensional but informative representation, a lower-variance linear model can generalise well from a small labelled dataset. More flexible models can capture non-linear interactions, but they also have more scope to fit validation-specific patterns. This also matches the lecture view of CNNs as neural networks that learn increasingly abstract features: the CNN performs the non-linear feature learning, while the final classifier only needs a simpler decision boundary.

The relative behaviour of Random Forest also differed across the two tasks. In Task 1, Logistic Regression was slightly stronger than Random Forest, suggesting that the coarse animal classes were already well separated in the EfficientNet feature space. In Task 2, Random Forest and Extra Trees achieved the best validation results, suggesting that fine-grained bird recognition benefited more from non-linear interactions between CNN dimensions and low-level descriptors. This conclusion should still be treated cautiously because the Task 2 validation split contains only 84 images; a difference of one or two predictions can noticeably change the reported accuracy. Overall, Task 1 favoured simpler linear classifiers once strong CNN features were available, whereas Task 2 benefited more from ensemble methods because the fine-grained bird classes required subtler non-linear feature interactions.

4.3 Error Analysis

The aggregate metrics suggest that Task 1 errors were relatively limited: the best Task 1 model reached 97.3% accuracy and 97.3% macro-F1. Because accuracy and macro-F1 were almost identical, the remaining mistakes were unlikely to be concentrated in one or two weak classes. This is consistent with the coarse-grained setting, where broad shape and appearance cues separate most animal categories. By

True label	Am. Goldfinch	Blue Jay	Cardinal	Herring Gull	House Sparrow	Red-winged Blackbird	Ring-billed Gull	Song Sparrow	Wilson Warbler	Yellow Warbler
Am. Goldfinch	8	0	0	0	0	0	0	0	0	0
Blue Jay	0	8	0	0	0	0	0	0	0	0
Cardinal	0	0	8	0	0	0	0	0	0	0
Herring Gull	0	0	0	9	0	0	0	0	0	0
House Sparrow	0	0	0	0	9	0	0	0	0	0
Red-winged Blackbird	0	0	0	0	0	9	0	0	0	0
Ring-billed Gull	0	0	0	1	0	0	7	0	0	0
Song Sparrow	0	0	0	0	0	0	0	9	0	0
Wilson Warbler	0	0	0	0	0	0	0	0	8	0
Yellow Warbler	0	0	0	0	0	0	0	0	1	7
	Am. Goldfinch	Blue Jay	Cardinal	Herring Gull	House Sparrow	Red-winged Blackbird	Ring-billed Gull	Song Sparrow	Wilson Warbler	Yellow Warbler
Predicted label										

Figure 1: Task 2 validation confusion matrix for Random Forest using HOG, additional features, and EfficientNet-V2-L embeddings.

contrast, Task 2 provides a clearer view of the remaining weakness of the approach because its classes are visually closer.

The Task 2 validation errors were concentrated in a small number of visually similar pairs. Using the HOG, additional features, and EfficientNet-V2-L representation, both Random Forest and Extra Trees made two errors on the 84-image validation split. In both models, one Ring-billed Gull image was predicted as Herring Gull, and one Yellow Warbler image was predicted as Wilson Warbler, as shown for Random Forest in Figure 1. The corresponding recalls were 7/8 for Ring-billed Gull and 7/8 for Yellow Warbler, while all other classes had perfect recall in these two models.

These errors are meaningful because they are not random confusions between unrelated species. Herring Gull and Ring-billed Gull share a similar white and grey body structure, so the distinction depends on subtle bill markings. Yellow Warbler and Wilson Warbler are also visually close because both are small yellow birds, and the distinguishing black cap of the Wilson Warbler may be hard to detect when the head region is small or poorly positioned. The absence of errors between visually distinctive species suggests that the model learned broad colour and shape cues, but still struggled when the decision depended on a small local visual cue.

4.4 Coarse-to-Fine Reflection

The coarse-to-fine progression reveals both the strength and limitation of the current approach. Transfer learning provides a strong general visual representation even when the labelled dataset is small, explaining why the same EfficientNet-based strategy worked well on both tasks. However, whole-image features are less ideal when the decision depends on a small discriminative part. Fine-grained classification therefore exposes a gap between recognising the general object type and localising the evidence needed to separate similar species.

4.5 Limitations

One limitation is that the pretrained CNNs were fixed feature extractors, so the representation was not adapted to the target bird species. The pipeline also did not perform object localisation, part detection, or region-of-interest modelling. Fine-grained recognition often benefits from focusing on the head, wing, or bill, but whole-image summaries may dilute these small cues (Zhang et al., 2014).

The small size of the Task 2 training set also makes model selection less stable. Although stratified cross-validation reduces dependence on a single split, the validation folds are still small. Therefore, small differences between the top models should not be over-interpreted. This is why I considered both accuracy and macro-F1 and compared several model families.

5 Conclusion

This project showed that image representation was the most important factor in both coarse-grained and fine-grained classification. The provided low-level descriptors were useful as baseline features, but the strongest results came from combining selected provided descriptors with ImageNet-pretrained CNN embeddings. In particular, the final models used EfficientNet-V2-L with HOG and additional features, plus colour for Task 1.

The coarse-to-fine comparison showed that fine-grained bird classification requires richer representations because the classes differ in subtle local details rather than broad visual categories. Once strong CNN embeddings were used, several classifiers achieved high accuracy, suggesting that the representation made the classes substantially more separable. Overall, the results support the value of transfer learning for small image classification datasets, while

also showing the limitations of whole-image features for fine-grained recognition.

6 AI Declaration

I declare that I used generative AI tools, including ChatGPT and Codex, to assist with project organisation, code structure, and report paraphrasing where appropriate. All experiments, results, analysis, and final report content were reviewed, verified, and submitted by me. No project materials or outputs were shared with any other student, staff member, or third party.

References

- Leo Breiman. 2001. Random forests. *Machine Learning*, 45(1):5–32.
- Corinna Cortes and Vladimir Vapnik. 1995. Support-vector networks. *Machine Learning*, 20(3):273–297.
- Navneet Dalal and Bill Triggs. 2005. Histograms of oriented gradients for human detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, volume 1, pages 886–893.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255.
- Pierre Geurts, Damien Ernst, and Louis Wehenkel. 2006. Extremely randomized trees. *Machine Learning*, 63(1):3–42.
- Mingxing Tan and Quoc V. Le. 2021. EfficientNetV2: Smaller models and faster training. In *Proceedings of the 38th International Conference on Machine Learning*, pages 10096–10106.
- Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. 2014. How transferable are features in deep neural networks? In *Advances in Neural Information Processing Systems*, volume 27.
- Matthew D. Zeiler and Rob Fergus. 2014. Visualizing and understanding convolutional networks. In *European Conference on Computer Vision*, pages 818–833.
- Ning Zhang, Jeff Donahue, Ross Girshick, and Trevor Darrell. 2014. Part-based R-CNNs for fine-grained category detection. In *European Conference on Computer Vision*, pages 834–849.